Process Plant Design and Economics (CHL4030)

Mini-Project Report

Design and Process Economic Evaluation of a 1000 TPD Methyl Acetate Plant

Esterification of Acetic Acid with Methanol

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Course	CHL4030: Process Plant Design and Economics
Submission Date	6 April 2026

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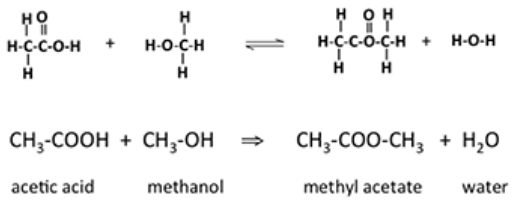
1 Introduction

This report covers the design and economic evaluation of a continuous 1000 TPD methyl acetate plant. The plant uses the liquid-phase esterification of acetic acid with methanol as its core reaction. **A plug flow reactor operating near 60 °C and 5 bar is used, followed by a three-column distillation train that recovers the product and recycles unreacted raw materials.**



The NRTL thermodynamic model was selected for simulation because the four-component system (acetic acid, methanol, methyl acetate, water) behaves in a strongly non-ideal way. The plug flow reactor was chosen because the reaction is positive-order and equilibrium-limited, and a PFR gives higher conversion per unit volume compared to a CSTR under similar conditions.

2 Project Scope and Design Basis

2.1 Design Basis

Most parameters are selected based on literature, while others are reasonable engineering assumptions.

- **Product:** methyl acetate
- **Plant capacity:** 1000 TPD = 1,000,000 kg/day
- **Continuous steady-state operation**
- **Operating days per year:** 330
- **Reactor temperature:** 60 °C
- **Reactor pressure:** $\approx$ 5 bar
- **Thermodynamic model:** NRTL
- **Acetic acid single-pass conversion:** 70%
- **Methanol-to-acetic acid molar feed ratio:** 2:1

2.2 Process Overview

The overall plant has four main sections:

- **Feed preparation** – static mixer, pump, and heater to condition the reactor feed.
- **Reaction** – isothermal plug flow reactor (Aspen `RPlug`).
- **Separation** – three RADFRAC columns to recover methyl acetate, recycle methanol, and

recycle acetic acid.

- **Recycle** – coolers and recycle loops returning unreacted material to the front of the plant.

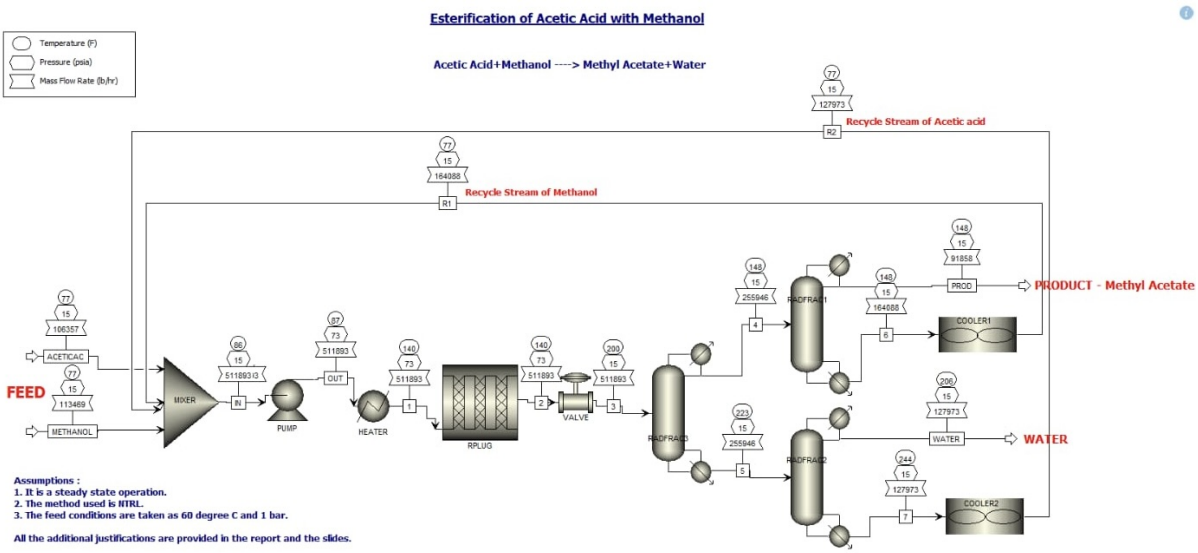


Figure 1: Aspen process flowsheet

3 Reaction, Thermodynamics, and Reactor Selection

3.1 The Reaction

Methyl acetate is produced by the reversible liquid-phase esterification:

$$CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O \tag{1}$$

The reaction is equilibrium-limited and is typically acid-catalysed. The liquid phase contains methanol, acetic acid, methyl acetate, and water, all of which interact strongly through hydrogen bonding. This non-ideal behaviour has to be captured properly in both the reactor and distillation calculations.

3.2 Why We Used the NRTL Thermodynamic Model

- The Non-Random Two Liquid (NRTL) model is a thermodynamic model used to accurately predict liquid-phase activity coefficients.
- It is particularly effective in representing strongly non-ideal mixtures, where molecular interactions deviate significantly from ideal behavior.
- The model accounts for specific intermolecular interactions, including hydrogen bonding between components.
- It is capable of reliably predicting azeotropic behavior in multicomponent systems.

For the methyl acetate system, NRTL gives a physically realistic description of both the reactor

environment and the downstream distillation columns.

3.3 Why We Chose a Plug Flow Reactor

The reactor was modelled as an Aspen `RPlug` block. The selection of reactor type and operating conditions was based on reaction kinetics and supported by literature.

- The esterification of acetic acid with methanol is a reversible, liquid-phase reaction with positive reaction order. For such systems, maximising conversion per unit reactor volume is critical.
- According to Fogler (*Elements of Chemical Reaction Engineering*), for reactions with positive reaction order, a plug flow reactor (PFR) achieves higher conversion than a CSTR for the same reactor volume. Therefore, a PFR is more suitable for equilibrium-limited reactions like esterification.
- Saha and Bhatia (2002) studied the kinetics of this reaction and confirmed that it occurs in the liquid phase, with optimal operating temperatures in the range of 55–70 °C. They also observed that higher conversions are achieved under plug flow behaviour.
- Industrial production of methyl acetate commonly employs packed-bed tubular reactors or reactive distillation columns. A packed-bed reactor closely approximates plug flow behaviour, justifying the use of the Aspen `RPlug` model.

4 Operating Conditions

The reactor conditions were chosen as shown in Table 1.

Table 1: Selected reactor operating conditions and reasons

Parameter	Value and reason
Temperature	60 °C. This is within the 55–70 °C range reported in literature and balances reaction rate against equilibrium limitations.
Pressure	≈5 bar. The process stays in the liquid phase, so pressure has little effect on equilibrium and is kept near atmospheric.
Phase	Liquid phase, giving good contact between reactants and straightforward temperature control.
Reactor type	Plug flow reactor (`RPlug`) for higher conversion per unit volume.

5 Material Balance Calculations

5.1 Basis and Molecular Weights

The design production rate is:

$$1000 \text{ TPD} = \frac{1{,}000{,}000 \text{ kg/day}}{24} = 41{,}666.67 \text{ kg/h} \tag{2}$$

Table 2 lists the molecular weights used.

Table 2: Molecular weights

Component	Formula	MW (kg/kmol)
Acetic acid	CH_3COOH	60.05
Methanol	CH_3OH	32.04
Methyl acetate	CH_3COOCH_3	74.08
Water	H_2O	18.00

The required molar production rate of methyl acetate is:

$$\dot{n}_{\text{MeOAc}} = \frac{41{,}666.67}{74.08} = 562.46 \text{ kmol/h} \tag{3}$$

5.2 Reactor Material Balance

Assuming 70% conversion of acetic acid:

$$\dot{n}_{A,\text{feed}} = \frac{562.46}{0.70} = 803.51 \text{ kmol/h} \tag{4}$$

With a 2:1 methanol-to-acetic acid molar ratio:

$$\dot{n}_{B,\text{feed}} = 2 \times 803.51 = 1607.01 \text{ kmol/h} \tag{5}$$

The reactor inlet and outlet follow directly from stoichiometry and are shown in Table 3.

Table 3: Reactor material balance (single pass, 70% acetic acid conversion)

Component	Feed (kmol/h)	Outlet (kmol/h)	Outlet (kg/h)	Note
Acetic acid	803.51	241.05	14,475	Unreacted; sent to recovery
Methanol	1607.01	1044.56	33,468	Unreacted; sent to recovery
Methyl acetate	0.00	562.46	41,667	Product
Water	0.00	562.46	10,124	By-product
Total	2410.52	2410.52	99,734	Mass conserved

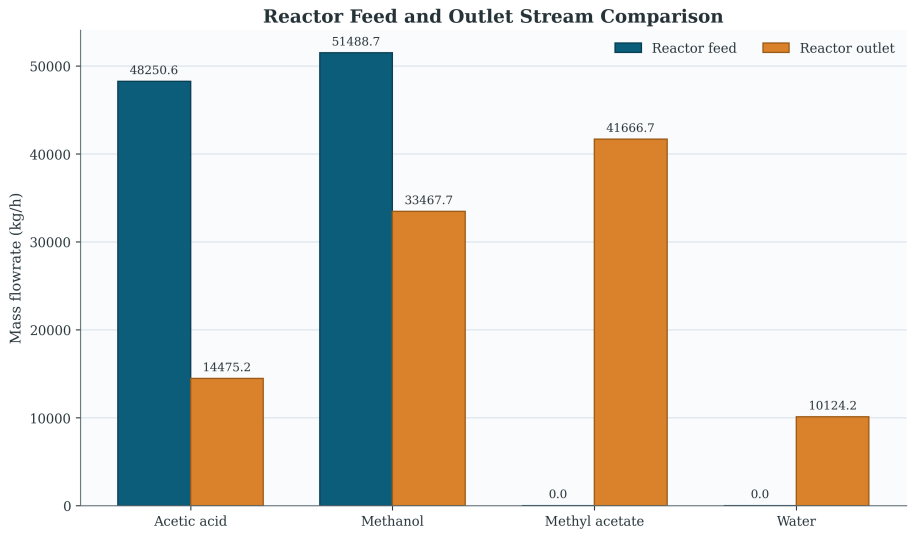


Figure 2: Reactor feed v/s outlet stream flow rates

5.3 Feed Preparation Section

The feed preparation train (mixer, pump, heater) does not involve any reaction or separation. Only pressure and temperature change. For each unit:

$$\dot{m}_{\text{in}} = \dot{m}_{\text{out}} \approx 99.73 \text{ t/h} \tag{6}$$

5.4 Separation Section

The three-column separation train processes the reactor outlet into product, recycle streams, and waste water:

- **RADFRAC 3** – first column; separates unreacted acetic acid for recycle, while heavier components (mainly water) proceed further.
- **RADFRAC 1** – second column; recovers methyl acetate as the overhead product stream.
- **RADFRAC 2** – third column; separates methanol for recycle, with water removed as the bottom product.

The resulting stream summary is given in Table 4.

Table 4: Separation train stream summary

Unit	Main output	Flowrate (kg/h)
RADFRAC 3 (overhead)	Acetic acid (recycle)	14,475
RADFRAC 3 (bottoms)	Water (waste/ purge)	10,124
RADFRAC 1 (overhead)	Methyl acetate (product)	41,667
RADFRAC 2 (overhead)	Methanol (recycle)	33,468

5.5 Overall Plant Balance

The recycle streams for methanol and acetic acid are internal to the plant and do not appear in the external balance. At steady state:

$$\text{Fresh feed} = \text{External products} + \text{External waste} \tag{7}$$

Table 5: Overall plant balance (recycle streams are internal)

Stream	Component	Flowrate (kg/h)
Fresh feed	Acetic acid	33,775
Fresh feed	Methanol	18,021
Product	Methyl acetate	41,667
Waste	Water	10,124

6 Aspen Plus Simulation

6.1 How the Simulation Was Built

The Aspen Plus model was built step by step:

1. **Feed preparation:** The acetic acid and methanol feed streams were introduced at 25 °C and 1 bar and then mixed to form a single feed stream.
2. **Pressurisation and heating:** The mixed stream was pressurised to 5 bar using a pump and subsequently heated to 60 °C to achieve the desired reactor inlet conditions.
3. **Reaction:** The conditioned feed was then sent to the reactor, which was modelled as an isothermal `RPlug` block.
4. **Depressurisation:** The reactor outlet stream was depressurised to 1 bar using a valve to ensure suitable operating conditions for downstream distillation columns. The pressure was reduced to facilitate vapor–liquid equilibrium and reduce energy requirements in the distillation columns.
5. **Separation:** The depressurised stream was fed to a series of `RADFRAC` distillation columns for separation, yielding approximately 1000 TPD of methyl acetate as the product.
6. **Cooling and recycle:** The unreacted methanol and acetic acid streams were cooled to approximately 25 °C using coolers and recycled back to the feed section.

This stepwise approach makes it easy to isolate convergence issues and compare each unit against the hand calculations.

6.2 Comparison with Hand Calculations

Table 6 compares the Aspen summary values with the hand-calculated results. The agreement is very close in all cases, confirming that the process assumptions are internally consistent.

Table 6: Aspen results vs. hand calculations

Quantity	Aspen	Hand calculations	Deviation
Feed section total flow	≈99,673 kg/h	99,734 kg/h	Negligible
Reactor outlet – acetic acid	≈14,473 kg/h	14,475 kg/h	Negligible
Reactor outlet – methanol	≈33,463 kg/h	33,468 kg/h	Negligible
Reactor outlet – methyl acetate	≈41,666 kg/h	41,667 kg/h	Negligible
Reactor outlet – water	≈10,124 kg/h	10,124 kg/h	Negligible
Product recovery (RADFRAC 1, overhead)	≈41,666 kg/h	41,667 kg/h	Negligible
Methanol recycle (RADFRAC 2, overhead)	≈33,463 kg/h	33,468 kg/h	Negligible
Acetic acid recycle (RADFRAC 3, overhead)	≈14,473 kg/h	14,475 kg/h	Negligible

7 Process Economic Calculations

7.1 Equipment Costs

The equipment costs were estimated using purchased cost charts from the book by Peters and Timmerhaus.

Table 7: Summary of purchased equipment costs

Equipment Description	Cost (USD)
Static mixers (3 units, stainless steel, $4,000 each)	12,000
Shell-and-tube heater (304 stainless steel, heat transfer area = 172 m^2)	25,000
Distillation columns: RADFRAC 1 (2.2 m dia), RADFRAC 2 (1.8 m dia), RADFRAC 3 (1.5 m dia), all stainless steel	41,000
Air cooler 1 (total area = 290 m^2, 3 units at $50,000 each)	150,000
Air cooler 2 (total area = 230 m^2, 3 units at $37,000 each)	111,000
Piping (100 m, 5-inch stainless steel, Schedule 80, $200/m)	20,000
Pipe insulation and painting	1,320
Control valve (5-inch)	3,000
Plug flow reactor (L = 13 m, D = 2 m, volume = 40.8 m^3, stainless steel)	250,000
Centrifugal pumps (1153 gal/min, stainless steel, 2 units at $9,000 each with 1.2 cost factor)	21,600
Total purchased equipment cost	**634,920**

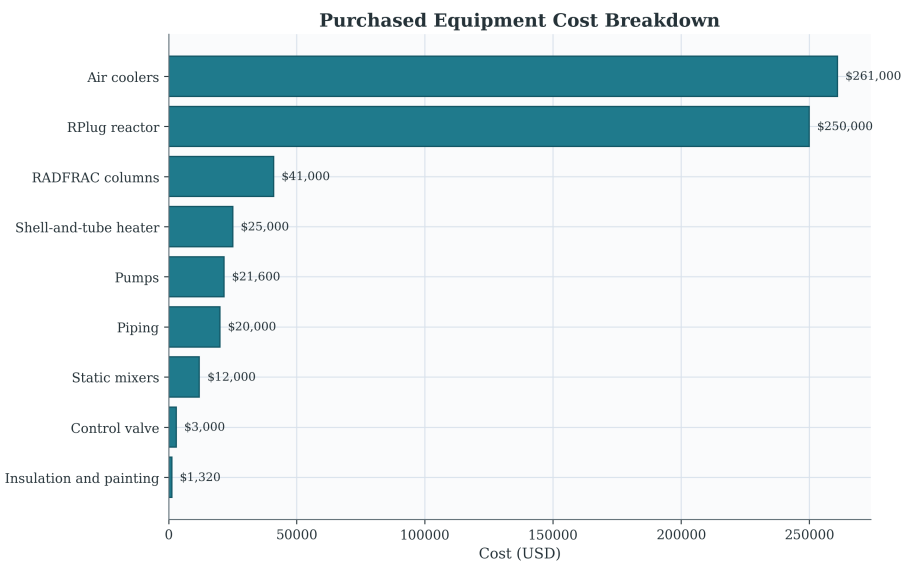


Figure 3: Equipment Cost Breakdown

7.2 Capital Cost Estimate

As per the following specified cost factors (all calculations on per year basis):

- Exchange rate: INR 83 per USD
- Delivery charges: 25% of purchased equipment cost
- Fixed capital investment (FCI): 5 $\times$ delivered equipment cost

Applying these factors:

$$\text{Purchased cost (INR)} = 634{,}920 \times 83 = 5{,}26{,}98{,}360 \text{ INR} = 5.27 \text{ crore} \tag{8}$$

$$\text{Delivered cost} = 1.25 \times 5.27 = 6.59 \text{ crore} \tag{9}$$

$$\text{FCI} = 5 \times 6.59 = 32.94 \text{ crore} \tag{10}$$

Table 8: Capital investment summary

Item	Value
Purchased equipment cost	5.27 crore INR
Delivered equipment cost	6.59 crore INR
Fixed capital investment (FCI)	32.94 crore INR

7.3 Revenue and Annual Operating Costs

Based on 1000 TPD production, a selling price of INR 40/kg, and 330 operating days per year:

$$\text{Annual revenue} = 1{,}000{,}000 \times 40 \times 330 = 1320 \text{ crore INR/year} \tag{11}$$

The annual operating costs were estimated using the class correlations and are listed in Table 9.

Table 9: Annual operating cost estimate

Cost item	Crore INR/year
Utilities (15% of TPC)	198.00
Raw materials (40% of TPC)	528.00
Operating labour	1.65
Operating supervision (15% of operating labour)	0.25
Maintenance and repairs (5% of FCI)	1.65
Plant overhead (60% of operating labour + supervision + maintenance)	2.13
Research and development (5% of TPC)	66.00
General expenses (20% of operating labour)	0.33
Distribution and marketing (10% of TPC)	132.00
Patents and royalties (6% of TPC)	79.20
Local property tax (3% of FCI)	0.99
Insurance (1% of FCI)	0.33
Total	**1010.52**

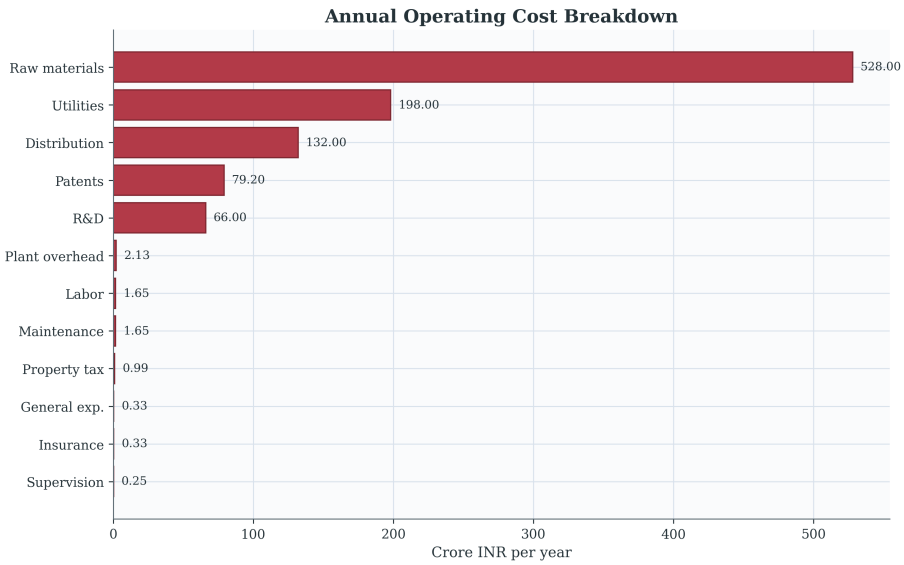


Figure 4: Annual Operating Cost Breakdown

7.4 Profitability and Payback Period

$$\text{Profit before depreciation} = 1320.00 - 1010.52 = 309.48 \text{ crore/year} \tag{12}$$

$$\text{Depreciation (10-year, straight line)} = \frac{32.94}{10} = 3.29 \text{ crore/year} \tag{13}$$

$$\text{Profit after depreciation} = 309.48 - 3.29 = 306.19 \text{ crore/year} \tag{14}$$

Annual cash flow (profit before depreciation, since depreciation is non-cash):

$$\text{Cash flow} = 309.48 \text{ crore/year} \tag{15}$$

$$\text{Payback period} = \frac{32.94}{309.48} = 0.106 \text{ years} \approx 39 \text{ days} \tag{16}$$

Table 10: Profitability summary

Metric	Value
Annual revenue	1320.00 crore INR/year
Total annual operating cost	1010.52 crore INR/year
Profit before depreciation	309.48 crore INR/year
Depreciation	3.29 crore INR/year
Profit after depreciation	306.19 crore INR/year
Annual cash flow	309.48 crore INR/year
Payback period	0.106 years ($\approx$ 39 days)

7.5 What the Numbers Really Mean

The payback period of about 39 days looks unrealistically short. This does not mean the real plant pays back in 39 days. It mainly reflects the simplified assumptions built into the assignment:

- Equipment costs are screening-level estimates.
- No cost escalation to a CEPCI base year was applied.
- Installation, contingency, and offsite costs were rolled into simple factors.
- The selling price and operating cost fractions come directly from class assumptions, not a current market study.

The economic results should therefore be treated as **preliminary** screening-level figures, not bankable projections.

8 Conclusions

1. A 1000 TPD methyl acetate plant based on esterification of acetic acid with methanol can be represented by a feed-preparation section, an isothermal PFR, and a three-column distillation train with recycle loops.
2. The NRTL thermodynamic model is appropriate because the system is strongly non-ideal and involves polar, associating components.
3. A plug flow reactor is the best choice for this project because of the positive-order, equilibrium-limited kinetics and because industrial methyl acetate plants use tubular or packed-bed reactors.
4. Operating at 60 °C, ≈5 bar, and a 2:1 methanol excess gives good reactor conversion and clean downstream separation.
5. Hand calculations and Aspen Plus results agree closely, confirming the flowsheet is internally consistent.
6. At the overall plant boundary, recycle streams are internal flows; only the stoichiometric fresh-feed requirement and the product and waste streams appear in the external balance.
7. Under the simplified class assumptions the plant appears economically attractive, but the result is highly sensitive to the equipment-cost basis and must be treated as a preliminary estimate.

9 Nomenclature

Table 11: Symbols and abbreviations

Symbol / term	Meaning
X_A	Conversion of acetic acid
$\dot{n}$	Molar flowrate (kmol/h)
$\dot{m}$	Mass flowrate (kg/h)
TPD	Tonnes per day
TPC	Total Product Cost (Revenue)
PFR	Plug flow reactor
RPlug	Aspen Plus plug flow reactor model
RADFRAC	Aspen Plus rigorous distillation column model
NRTL	Non-Random Two-Liquid thermodynamic model
FCI	Fixed capital investment
TCI	Total capital investment
SS	Stainless steel
CEPCI	Chemical Engineering Plant Cost Index

10 References

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